

TENSOR TRIANGULAR CHOW GROUPS

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ABSTRACT. I propose a definition of the Chow group of a rigid tensor triangulated category. One of the key ideas is to allow “generalized” cycles, with non-integral coefficients. The proposed relations might need some fine-tuning.

Hypothesis 1. Let $\mathcal{K}$ be an essentially small tensor triangulated category. Let us assume that its triangular spectrum in the sense of [1], $\mathrm{Spc}(\mathcal{K}) = \{ \mathcal{P} \subset \mathcal{K} \mid \mathcal{P} \text{ is prime} \}$, is a *noetherian* topological space, i.e. that every open of $\mathrm{Spc}(\mathcal{K})$ is quasi-compact. Let us also assume that $\mathcal{K}$ is *rigid*, as explained in [4] (or [2], where this property was called *strongly closed*). These hypotheses allow us to use the techniques of filtration of $\mathcal{K}$ by (generalized) dimension of the support.

Definition 2. As in [2, Def.3.1], let us consider $\dim : \mathrm{Spc}(\mathcal{K}) \longrightarrow \mathbb{Z} \cup \{\pm\infty\}$ a *dimension function*, meaning that $\mathcal{P} \subseteq \mathcal{Q} \implies \dim(\mathcal{P}) \leq \dim(\mathcal{Q})$, with equality in the finite range only if $\mathcal{P} = \mathcal{Q}$ (i.e. $\mathcal{P} \subseteq \mathcal{Q}$ and $\dim(\mathcal{P}) = \dim(\mathcal{Q}) \in \mathbb{Z}$ forces $\mathcal{P} = \mathcal{Q}$). Examples are the Krull dimension of $\overline{\{\mathcal{P}\}}$, or the opposite of its Krull codimension in $\mathrm{Spc}(\mathcal{K})$. Assuming $\dim(-)$ is clear from the context, we shall use the notation

$$\mathrm{Spc}(\mathcal{K})_{(p)} := \{ \mathcal{P} \in \mathrm{Spc}(\mathcal{K}) \mid \dim(\mathcal{P}) = p \}.$$

Remark 3. In my opinion, there is nothing conceptually remarkable about the free abelian group on $\mathrm{Spc}(\mathcal{K})_{(p)}$. Therefore I propose another definition of p -dimensional cycles. This requires some preparation.

Definition 4. Recall from [3, §4] that a rigid tensor triangulated category $\mathcal{L}$ is called *local* if $a \otimes b = 0 \implies a = 0$ or $b = 0$. This means that $\mathrm{Spc}(\mathcal{L})$ has a unique closed point $* := 0 \subset \mathcal{L}$, which is prime by assumption.

Example 5. For every tensor triangulated category $\mathcal{K}$ and for every prime $\mathcal{P} \in \mathrm{Spc}(\mathcal{K})$, the following tensor triangulated category is local in the above sense:

$$\mathcal{K}_{\mathcal{P}} := (\mathcal{K}/\mathcal{P})^{\natural}.$$

We call $\mathcal{K}_{\mathcal{P}}$ the *local category at $\mathcal{P}$* . There is an obvious (localization) functor

$$q_{\mathcal{P}} : \mathcal{K} \twoheadrightarrow \mathcal{K}/\mathcal{P} \hookrightarrow \mathcal{K}_{\mathcal{P}},$$

which is the composition of Verdier localization and idempotent completion. (The category $\mathcal{K}_{\mathcal{P}}$ is also the colimit of the $\mathcal{K}(U)$ over those (quasi-compact) open subsets $U \subseteq \mathrm{Spc}(\mathcal{K})$ which contain $\mathcal{P}$. See more in [4, §2.2] if necessary.)

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Definition 6. Assuming that $\mathcal{L}$ is local and that $\mathrm{Spc}(\mathcal{L})$ is noetherian, the open complement of the unique closed point $\{*\}$ in $\mathrm{Spc}(\mathcal{L})$ is quasi-compact, i.e. $\{*\}$ is a “Thomason (closed) subset”. Under the classification of thick $\otimes$ -ideals of $\mathcal{L}$, see [1], this one-point subset corresponds to the minimal non-zero subcategory

$$\mathrm{Min}(\mathcal{L}) := \mathcal{L}_{\{*\}} = \{a \in \mathcal{L} \mid \mathrm{supp}(a) \subseteq \{*\}\}.$$

These are the objects with minimal possible support (empty or a point).

Remark 7. Some comments are in order :

- (1) This subcategory was called the subcategory of *finite-length* objects in [2] and denoted $\mathrm{FL}(\mathcal{L})$. As far as I know, there is no reason for objects of $\mathrm{Min}(\mathcal{L})$ to have finite-length (in the categorical sense that they admit a finite filtration with simple subquotients). The present notation, $\mathrm{Min}(\mathcal{K})$, is less biased towards commutative algebra and therefore probably preferable.
- (2) As the previous comment suggests, if we take $\mathcal{L} = \mathrm{K}^b(R\text{-proj})$ the category of perfect complexes for R noetherian and local, then $\mathcal{L}$ is local and $\mathrm{Min}(\mathcal{L})$ is the subcategory of perfect complexes with finite-length homology.
- (3) One can of course consider $\mathrm{Min}(\mathcal{L})$ even if $*$ is not Thomason but in that case it would just be the zero subcategory $0 = \mathcal{L}_\emptyset$.

Definition 8. Let $p \in \mathbb{Z}$. Then I define the group of *generalized p -cycles* to be

$$\mathrm{Z}_p(\mathcal{K}) := \bigoplus_{\mathcal{P} \in \mathrm{Spc}(\mathcal{K})_{(p)}} K_0(\mathrm{Min}(\mathcal{K}_{\mathcal{P}})),$$

where K_0 is the Grothendieck K -group (the quotient of the monoid of isomorphism classes $[a]$ of objects under $\oplus$, by the submonoid of the $[a] + [\Sigma b] + [c]$ for which there exists a distinguished triangle $a \rightarrow b \rightarrow c \rightarrow \Sigma a$).

Out of nostalgia for usual cycles, a generalized p -cycle can be written $\sum_{\mathcal{P}} \lambda_{\mathcal{P}} \cdot \mathcal{P}$ or $\sum_{\mathcal{P}} \lambda_{\mathcal{P}} \cdot \overline{\{\mathcal{P}\}}$, which is a purely notational choice. The non-trivial point is that we allow coefficients $\lambda_{\mathcal{P}}$ to live in other abelian groups than $\mathbb{Z}$, namely the Grothendieck groups of the minimal categories at every $\mathcal{P}$.

Example 9. Let X be a (topologically) noetherian scheme and $\mathcal{K} = \mathrm{D}^{\mathrm{perf}}(X)$ the derived category of perfect complexes, whose spectrum $\mathrm{Spc}(\mathcal{K}) \cong X$ recovers the underlying space of X . Let $\dim(-)$ be the (opposite of the) Krull (co)dimension. Then we recover the usual p -dimensional (resp. $(-p)$ -codimensional) cycles. Indeed, we have by Thomason that $\mathcal{K}_{\mathcal{P}} \cong \mathrm{K}^b(\mathcal{O}_{X,x}\text{-proj})$ if $\mathcal{P} \in \mathrm{Spc}(\mathcal{K})$ corresponds to $x \in X$. The reason why integral coefficients suffice over regular schemes is that the group homomorphism defined by length of homology

$$K_0(\mathrm{Min}(\mathrm{K}^b(\mathcal{O}_{X,x}\text{-proj}))) \longrightarrow \mathbb{Z},$$

is an isomorphism if X is regular (at x). However, in general, the left-hand group could be tricky. It is discussed for instance in [5].

Now to the relations. For this, we need a preparation.

Lemma 10. *Recall Hypothesis 1. Let $\mathcal{Q} \in \mathrm{Spc}(\mathcal{K})$ be a prime with $\dim(\mathcal{Q}) \in \mathbb{Z}$ finite. Let $a \in \mathcal{K}$ be an object such that $\mathrm{supp}(a) = \overline{\{\mathcal{Q}\}}$. Let $\alpha : a \rightarrow a$ be an endomorphism such that $\mathcal{Q} \notin \mathrm{supp}(\mathrm{cone}(\alpha))$, meaning that $\mathrm{cone}(\alpha) \in \mathcal{Q}$, or equivalently that $\alpha : a \rightarrow a$ is an isomorphism at $\mathcal{Q}$, i.e. in $\mathcal{K}_{\mathcal{Q}}$. Then there exists finitely many $\mathcal{P} \in \mathrm{supp}(\mathrm{cone}(\alpha))$ with $\dim(\mathcal{Q}) = \dim(\mathcal{P}) - 1$. These $\mathcal{P}$ all are*

generic points of some irreducible components of $\text{supp}(\text{cone}(\alpha))$. Moreover, for every such $\mathcal{P}$, we have $q_{\mathcal{P}}(\text{cone}(\alpha)) \in \text{Min}(\mathcal{K}_{\mathcal{P}})$ where $q_{\mathcal{P}} : \mathcal{K} \rightarrow \mathcal{K}_{\mathcal{P}}$.

Proof. Let $Z := \text{supp}(\text{cone}(\alpha))$. Since $Z \subseteq \text{supp}(a) = \overline{\{\mathcal{Q}\}}$, every $\mathcal{P} \in Z$ belongs to $\overline{\{\mathcal{Q}\}}$, meaning $\mathcal{P} \subseteq \mathcal{Q}$ and in particular $\dim(\mathcal{P}) \leq \dim(\mathcal{Q})$. Since $\mathcal{P} = \mathcal{Q}$ is excluded by hypothesis, we must even have $\dim(\mathcal{P}) \leq \dim(\mathcal{Q}) - 1$ by Definition 2. Moreover, since $\text{Spc}(\mathcal{K})$ is noetherian and spectral, there are finitely many $\mathcal{P}_1, \dots, \mathcal{P}_n \in Z$ such that $Z = \overline{\{\mathcal{P}_1\}} \cup \dots \cup \overline{\{\mathcal{P}_n\}}$ is the decomposition in irreducible components. By the previous discussion $\dim(\mathcal{P}_i) \leq \dim(\mathcal{Q}) - 1$. Now, if $\mathcal{P} \in Z$ satisfies $\dim(\mathcal{P}) = \dim(\mathcal{Q}) - 1$, it must belong to $\overline{\{\mathcal{P}_i\}}$ for some i , meaning $\mathcal{P} \subseteq \mathcal{P}_i$. This yields $\dim(\mathcal{Q}) - 1 = \dim(\mathcal{P}) \leq \dim(\mathcal{P}_i) \leq \dim(\mathcal{Q}) - 1$ which forces equality (of dimensions) and therefore equality of primes $\mathcal{P} = \mathcal{P}_i$ by Definition 2 again.

For the moreover part, let $\mathcal{P} \in \text{supp}(\text{cone}(\alpha))$ be such that $\dim(\mathcal{P}) = \dim(\mathcal{Q}) - 1$. Recall from [1, Prop. 3.6] that for any tensor triangular functor $F : \mathcal{K} \rightarrow \mathcal{L}$, we have $\text{supp}_{\mathcal{L}}(F(b)) = \varphi^{-1}(\text{supp}_{\mathcal{K}}(b))$ for every $b \in \mathcal{K}$, where $\varphi = \text{Spc}(F) : \text{Spc}(\mathcal{L}) \rightarrow \text{Spc}(\mathcal{K})$ is the induced map on spectra. Also recall from [1, Prop. 3.11 and Cor. 3.14] that for $F = q_{\mathcal{P}} : \mathcal{K} \rightarrow \mathcal{K}_{\mathcal{P}}$, the map $\text{Spc}(q_{\mathcal{P}})$ is a homeomorphism of $\text{Spc}(\mathcal{K}_{\mathcal{P}})$ onto the subspace $\{\mathcal{P}' \in \mathcal{K} \mid \mathcal{P}' \supseteq \mathcal{P}\}$ of $\text{Spc}(\mathcal{K})$. Combining those two, we can identify

$$\text{supp}_{\mathcal{K}_{\mathcal{P}}}(q_{\mathcal{P}}(b)) = \text{supp}_{\mathcal{K}}(b) \cap \{\mathcal{P}' \in \text{Spc}(\mathcal{K}) \mid \mathcal{P}' \supseteq \mathcal{P}\}$$

for every $b \in \mathcal{K}$. Applying this to $b = \text{cone}(\alpha)$ shows that every point $\mathcal{P}' \in \text{supp}_{\mathcal{K}_{\mathcal{P}}}(q_{\mathcal{P}}(\text{cone}(\alpha)))$ can be identified with some $\mathcal{P}' \in \text{supp}_{\mathcal{K}}(\text{cone}(\alpha)) = Z$ such that $\mathcal{P}' \supseteq \mathcal{P}$, meaning that $\mathcal{P} \in \overline{\{\mathcal{P}'\}}$. But we have seen that $\mathcal{P}$ is the generic point of some irreducible component of Z , so $\mathcal{P} \in \overline{\{\mathcal{P}'\}} \subseteq Z$ forces $\mathcal{P}' = \mathcal{P}$. Back in $\text{Spc}(\mathcal{K}_{\mathcal{P}})$, where $\mathcal{P}' = \mathcal{P}$ corresponds to the unique closed point $*$, we then get $\text{supp}_{\mathcal{K}_{\mathcal{P}}}(q_{\mathcal{P}}(\text{cone}(\alpha))) \subseteq \{*\}$ as wanted. $\square$

Definition 11. Let $p \in \mathbb{Z}$ and let $\mathcal{Q} \in \text{Spc}(\mathcal{K})_{(p+1)}$. Let $a \in \mathcal{K}$ such that $\text{supp}(a) = \overline{\{\mathcal{Q}\}}$ and $\alpha : a \rightarrow a$ such that $\text{cone}(\alpha) \in \mathcal{Q}$. Let us denote by

$$\text{div}(\alpha) := \sum_{\mathcal{P} \in \text{Spc}(\mathcal{K})_{(p)}} [q_{\mathcal{P}}(\text{cone}(\alpha))] \cdot \mathcal{P}.$$

By Lemma 10, this is a well-defined element of $Z_p(\mathcal{K})$.

I define the subgroup of p -boundaries $B_p(\mathcal{K})$ inside the group $Z_p(\mathcal{K})$ of p -cycles, as the subgroup generated by the following subset

$$\{ \text{div}(\alpha) \mid (\mathcal{Q}, a, \alpha) \text{ as above} \}$$

which belongs to $Z_p(\mathcal{K})$ by the above Lemma.

Finally, I define the p -dimensional “Chow” group of $\mathcal{K}$ as the quotient

$$\text{CH}_p(\mathcal{K}) = Z_p(\mathcal{K}) / B_p(\mathcal{K}).$$

Remark 12. There is a possible variant for the relations, where one would drop $\text{supp}(a) = \overline{\{\mathcal{Q}\}}$ and allow endomorphisms $\alpha : a \rightarrow a$ to be any endomorphism such that $\dim(\text{supp}(a)) = p + 1$ and $\text{supp}(\text{cone}(\alpha)) = p + 1$, where the dimension is extended to closed subsets by $\dim(Z) = \sup_{\mathcal{P} \in Z} \dim(\mathcal{P})$ as usual. A lemma very similar to Lemma 10 can be applied then.

Remark 13. I am under the impression that both definitions coincide for $\mathcal{K} = \text{D}^{\text{perf}}(X)$ with X regular (say, smooth projective). Only further investigation will show what is the best relation, hopefully one of the above.

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